

Detecting Clickbait Headlines with Traditional Machine Learning and Transformer Models

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Abstract

Clickbait headlines have become increasingly prevalent in online news and social media, often prioritizing user engagement over informational quality. This phenomenon negatively impacts user trust and content credibility, motivating the need for automatic clickbait detection systems. In this paper, we address clickbait detection as a binary text classification task and present a comparative study between a classical machine learning approach and a modern transformer-based model.

We implement a Support Vector Machine (SVM) classifier that combines TF-IDF representations with handcrafted lexical features specifically designed to capture stylistic patterns common in clickbait headlines. In parallel, we propose a transformer-based model built on RoBERTa, enhanced with an attention-based pooling mechanism to improve sequence-level representation and classification performance.

Both models are evaluated on a balanced dataset of 32,000 news headlines using standard classification metrics, including precision, recall, macro-F1 score, accuracy, and PR-AUC. Experimental results show that while the SVM model provides strong precision and serves as a reliable baseline, it exhibits lower recall for the clickbait class. In contrast, the RoBERTa-based model achieves consistently superior and more balanced performance across all metrics, demonstrating its effectiveness in capturing subtle contextual and semantic cues.

The findings highlight the advantages of transformer-based approaches for clickbait detection and underline their suitability for real-world deployment in dynamic online media environments.

1 Introduction

In the past 10 years, clickbait headlines have significantly increased due to the rapid growth of online news platforms and social media that have led to an overwhelming volume of digital content, where headlines play one of the most important roles in capturing the user's attention. Because of the competitive environment publishers employ clickbait headlines to encourage users to click on an article, often at the expense of informational quality and accuracy.

Due to the negative aspects of clickbait, as a result, the automatic detection of clickbait headlines has become an important research problem, particularly within the field of Natural Language Processing (NLP), where it is commonly addressed as a binary text classification task. Clickbait detection poses several challenges for NLP systems, as such headlines often rely on subtle linguistic cues, stylistic patterns, and implicit semantics rather than explicit keywords. Traditional machine learning approaches, such as Support Vector Machines (SVM),

have demonstrated strong performance when combined with carefully engineered features, including lexical statistics and term-weighting techniques. More recently, transformer-based models, such as RoBERTa, have achieved state-of-the-art results across a wide range of NLP tasks by leveraging deep contextual representations of text.

From a real scenario, a reliable clickbait detection system can be integrated into platforms and social media applications, helping to improve content quality and enhance the user’s overall experience.

The main objective is to create and compare the effectiveness of a traditional machine learning approach and a modern transformer-based model for clickbait detection. We will evaluate a Support Vector Machine (SVM) classifier trained on a custom pipeline and a RoBERTa model that was fine-tuned for binary classification.

The models are engineered by design of a feature-rich SVM-based clickbait detection model that combines lexical and statistical text representations, as long with the implementations customized RoBERTa architecture enhanced with an attention mechanism for headline classification. Therefore a comprehensive experimental comparison of the two approaches using evaluating metrics, highlighting their strengths and limitations in the context of clickbait detection.

1.1 Structure

The remainder of this paper is structured as follows. **Related Work** reviews related work on clickbait detection, covering both classical machine learning approaches and transformer-based models. **Corpus and Data Preprocessing** describes the dataset and the preprocessing steps applied to the text data. **Methodology and Models implementation** presents the proposed methodologies, detailing the SVM-based pipeline and the RoBERTa-based model architecture. **Results and Evaluation** reports the experimental setup and evaluation results, including a comparative analysis and error analysis. **Critical Analysis** provides a critical discussion of the findings, outlining limitations and potential sources of bias. Finally, **Conclusion** concludes the paper and discusses directions for future work.

2 Related Work

2.1 Fundamental NLP approaches

2.1.1 Machine Learning Algorithms

Support vector machines (SVMs) are a set of related supervised learning methods used for classification and regression. They belong to a family of generalized linear classifiers [4]. In another terms, Support Vector Machine (SVM) is a classification and regression prediction tool that uses machine learning theory to maximize predictive accuracy while automatically avoiding over-fit to the data [4]. Support Vector Machines (SVMs) are learning systems that use linear functions in a high-dimensional feature space and are trained using optimization techniques grounded in statistical learning theory. [4] They gained prominence after achieving performance comparable to advanced neural networks in handwriting recognition tasks using pixel-based inputs. The theoretical foundations of SVMs were developed by Vapnik and are based on the Structural Risk Minimization (SRM) principle, which minimizes an upper bound on the expected risk and provides better generalization than the Empirical Risk Minimization (ERM) approach used in traditional neural networks. [4] Initially designed for classification,

SVMs have since been extended to regression and are widely applied in pattern recognition tasks such as handwriting and face analysis [4].

In computer science, the problem of classifying false and misleading content have existed for so long and various attempts to solve or at least classify such content have been made in various applications using SVM approach from spam detection to fake news detection [5].

An experiment was conducted where the dataset contained 22,033 posts and was divided to 2495 post for training the model, and 19538 posts used for the validation of the model [5]. Generally, Extraction of the features is a process that relies heavily on domain experience and previous researches that proved the functioning of certain features. This process is the most critical part in model training. This is due to the fact that if the features extracted have low correlation with the labels, no matter what modeling algorithm, the model will fail to classify correctly. Initially, we extracted 28 features [5]. Initially, 28 features were extracted. However, increasing the number of features raises computational cost and may negatively impact model performance. Highly correlated or redundant features can reduce accuracy, and high-dimensional models require more data to ensure reliable learning. To address these issues, recursive feature elimination (RFE) was applied to reduce dimensionality. RFE iteratively removes less relevant features while considering feature interactions, resulting in the elimination of four features [5]. Although feature elimination reduced dimensionality, the data remained high-dimensional, which can negatively affect some learning algorithms. Therefore, Support Vector Machines (SVMs) were chosen due to their effectiveness in high-dimensional spaces. SVM represents data points as vectors in a high-dimensional space and separates classes using an optimal hyperplane, which may be linear or non-linear depending on data linearity. A linear SVM performed best, indicating that the data is largely linearly separable. The hyperplane maximizes the margin between classes and the nearest data points, providing robustness to noise and high-dimensional data [5]. A linear Support Vector Machine (SVM) was employed due to its effectiveness in high-dimensional and noisy data environments. The SVM achieved an accuracy of approximately 79% on the validation set, with precision, recall, and F1-score close to 0.79 [5].

SVM was selected for its margin-based classification approach, which maximizes the distance between classes using only the most informative data points near the decision boundary. This makes SVM more robust to noise and better suited for web-based data, where content style may evolve over time. Model parameters, including margin size and linearity, were optimized using Grid Search [5]. Additionally, SVM demonstrated strong generalization ability, achieving a training accuracy of 74.5% and a higher validation accuracy of 79.4%, reflecting the difference in data distribution between training and validation sets [5].

This research proposed an alternative approach to clickbait classification by emphasizing the importance of social media post elements and introducing effective features. The results demonstrate that clickbait detection can be successfully performed using a limited number of features, achieving reliable performance as measured by standard binary classification metrics. This confirms the feasibility of building efficient, real-time clickbait classifiers for social media platforms [5].

2.2 Transformer-based models

BERT (Bidirectional Encoder Representations from Transformers) has significantly advanced Natural Language Processing (NLP) by improving the performance of language models across diverse tasks. This review examines BERT's architecture, its applications in multiple NLP tasks, and extensions of its original design [3]. The study analyzes methodological aspects

of existing research and evaluates BERT’s impact on tasks such as named entity recognition, part-of-speech tagging, dependency parsing, question answering, machine translation, sentiment analysis, fake review detection, and cross-lingual transfer learning [3]. By synthesizing findings from prior work, this review highlights current challenges and future directions of BERT-based models, aiming to support both new and experienced NLP researchers and to encourage further innovation in language model development [3]. BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based language model that achieves strong performance across a wide range of NLP tasks due to its bidirectional training and masked language modeling [3]. By capturing contextual information from both directions, BERT significantly improves tasks such as question answering and linguistic inference [3]. Its architecture consists of stacked transformer encoder layers with self-attention mechanisms (Ghojogh and Ghodsi 2020), enabling effective contextual representation learning [3].

RoBERTa is an improved pretraining approach derived from BERT that addresses limitations in BERT’s original training procedure. It demonstrates that BERT was undertrained and shows that performance can be significantly improved through optimized training strategies [6]. RoBERTa achieves superior results by training on larger datasets with bigger batch sizes and longer sequences, removing the next sentence prediction objective, and dynamically varying the masking patterns during training. Additionally, it leverages large-scale datasets such as CC-NEWS, enabling it to match or outperform existing post-BERT models across multiple NLP benchmarks [6].

2.3 Evaluation of SVM versus BERT for Clickbait Detection

An experiment was conducted on the detection of Clickbait headlines using machine learning algorithms and transformer-based models where the datasets included various online news outlets. The dataset consists of 32,000 articles labeled as clickbait or non-clickbait, collected from multiple news and media websites, including BuzzFeed, Viral Nova, Upworthy, Wiki News, The New York Times, The Hindu, and The Guardian [1].

Support Vector Machine (SVM) is a supervised learning algorithm commonly applied to classification problems. It aims to separate data points from different classes by learning an optimal hyperplane that maximizes the margin between them. For unseen data, SVM predicts class labels based on their position relative to the decision boundary. Depending on the data distribution, SVM can be implemented using linear or non-linear kernels, enabling effective classification in both simple and complex feature spaces [1].

Transfer learning has gained significant attention in recent years, enabling knowledge learned from one task to be applied to related problems. While widely successful in computer vision, its adoption in natural language processing remained limited until the introduction of BERT (Bidirectional Encoder Representations from Transformers). BERT leverages transformer-based encoders with self-attention mechanisms to capture contextual relationships by processing entire sentences bidirectionally, rather than sequentially. This capability makes BERT an effective transfer learning model for a wide range of NLP tasks [1].

Accuracy, precision, recall, and F1-score were used in this study to evaluate the performance of the selected techniques. These metrics are derived from the confusion matrix, which summarizes the correctly and incorrectly classified instances for a classification problem, providing a clear understanding of model performance [1].

The study shows that BERT achieved the highest **accuracy** (0.977), outperforming other algorithms by about 3% on average. Support Vector Machines (SVM) delivered the second-best performance, with an accuracy of 0.97.

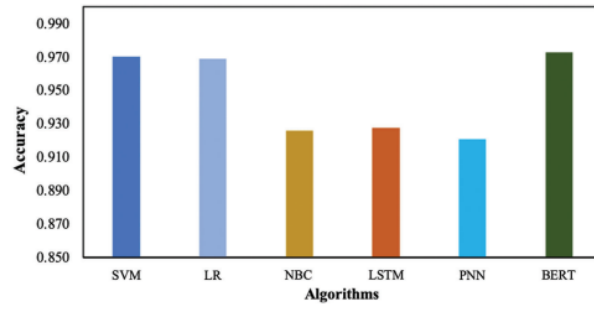


Figure 1: Accuracy achieved by learning algorithms [1].

BERT and SVM were the top-performing algorithms in terms of **precision**, each achieving a score of 0.979, slightly above the average of the other methods.

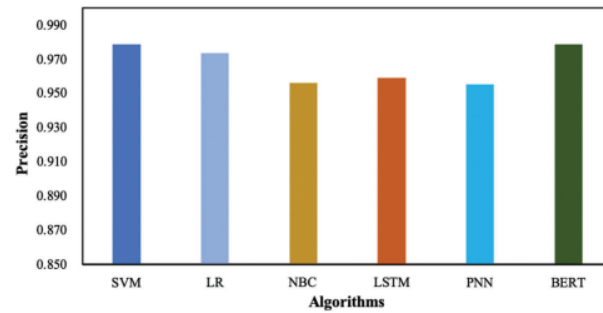


Figure 2: Precision achieved by learning algorithms [1].

BERT achieved the second-highest **recall** among the evaluated algorithms, with a score less than 1% below the best-performing logistic regression while SVM was on the third place.

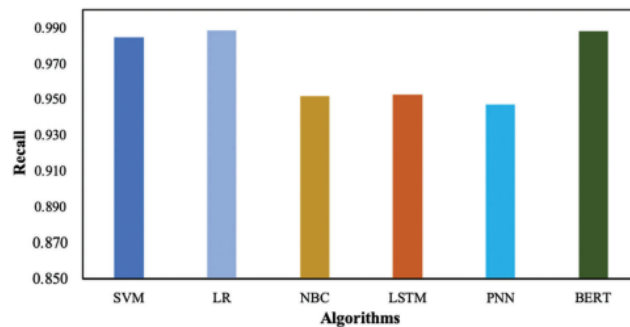


Figure 3: Recall achieved by learning algorithms [1].

BERT achieved the highest **F1-score** (0.983), followed closely by SVM with 0.98, demonstrating strong overall performance across evaluation metrics.

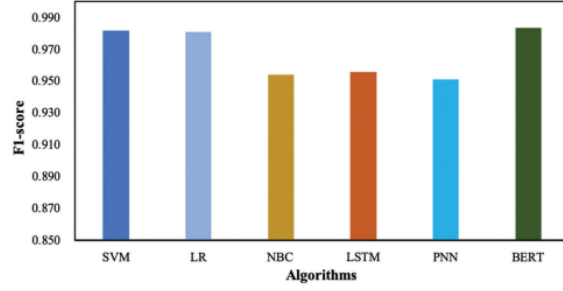


Figure 4: F1-score achieved by learning algorithms [1].

Clickbait detection is a growing challenge on social media, affecting both users and platforms. In this study, several machine learning algorithms were evaluated, with BERT emerging as the best-performing model across multiple evaluation metrics. While some algorithms, such as PNN, showed weaker performance, BERT demonstrated strong potential for accurate automated clickbait detection. Future work should explore the generalization of these models to datasets in other languages with different syntactic and semantic structures.

2.4 Limitations

Support Vector Machines (SVMs) face several challenges when applied to clickbait detection. First, they rely heavily on feature extraction, such as TF-IDF, n-grams, or precomputed embeddings, and cannot process raw text directly. This makes it difficult to capture subtle linguistic cues, contextual semantics, or curiosity-inducing patterns typical of clickbait, which may result in misclassification of superficially similar headlines [8]. Second, SVMs struggle with complex language structures; while kernel methods allow some non-linearity, highly contextual and sequential patterns are better handled by deep learning models like LSTMs or transformers. Third, SVM training can be computationally expensive, particularly on large datasets or with non-linear kernels, limiting scalability for real-time detection and streaming data scenarios. Fourth, clickbait datasets are often imbalanced, with far fewer clickbait examples than non-clickbait, making SVMs sensitive to class imbalance and prone to biased predictions unless class weighting or oversampling techniques are applied. Fifth, interpretability is limited: linear SVMs are somewhat explainable, but non-linear kernels obscure the decision-making process, which may be an issue for applications requiring transparency [8]. Finally, SVMs have limited adaptability; they cannot easily leverage contextual embeddings like BERT without precomputing features, and evolving clickbait styles require retraining the model to maintain accuracy. These limitations highlight the need for careful feature engineering, preprocessing, and consideration of alternative models for robust clickbait detection [8].

Despite its strong performance, **BERT** has several limitations for clickbait detection. Its large size leads to high computational costs and latency, making real-time deployment challenging. Fine-tuning requires substantial labeled data, and evolving clickbait styles necessitate frequent retraining. BERT is also difficult to interpret, sensitive to domain shifts, and primarily relies on textual features, often overlooking contextual signals such as images or user metadata [7].

2.5 Original Contribution

This project introduces several novel aspects to clickbait detection:

- **Custom TF-IDF vectorization:** Our TF-IDF implementation penalizes non-informative words and boosts words indicative of clickbait, enhancing feature relevance for classification.
- **Lexicon-based features:** We define a personalized lexicon of common clickbait and non-clickbait words, extracting features such as word counts, title length, and punctuation usage, providing additional semantic cues.
- **Hybrid SVM model:** The combination of custom TF-IDF and lexicon features improves SVM classification, making it robust to ambiguous or short headlines.
- **Customized RoBERTa classifier:** Incorporates dropout, a fully connected layer, and optional attention mechanisms to highlight influential tokens, improving deep learning performance and interpretability.
- **Ensemble approach:** Combines SVM and RoBERTa predictions, yielding a more stable and accurate clickbait detection system.

3 Corpus and Data Preprocessing

3.1 Corpus description

This dataset contains headlines from various news sites such as ‘WikiNews’, ‘New York Times’, ‘The Guardian’, ‘The Hindu’, ‘BuzzFeed’, ‘Upworthy’, ‘ViralNova’, ‘Thatscoop’, ‘Scoop-whoop’ and ‘ViralStories’. It has two columns first one contains headlines and the second one has numerical labels of clickbait in which 1 represents that it is clickbait and 0 represents that it is non-clickbait headline. The dataset contains total 32000 rows of which 50% are clickbait and other 50% are non-clickbait [2]. After data preprocessing a new column was created with the preprocessed text, where the data was splitted into train 80% (25600), validation 15% (4800) and test 5% (1600).

3.2 Text Preprocessing

The text preprocessing pipeline was designed to reduce noise and ensure a consistent representation of the headlines before model training. First, all text was converted to lowercase in order to avoid treating words with different capitalization as distinct tokens.

Next, non-alphanumeric characters such as punctuation marks were removed using regular expressions. The cleaned text was then tokenized using word-level tokenization. Stopwords were eliminated to remove common function words that do not contribute significant semantic information to the task of clickbait detection.

After stopword removal, lemmatization was applied. Lemmatization reduces words to their base or dictionary form, allowing different inflected forms of a word to be treated as a single feature. Finally, the processed tokens were rejoined into a single normalized text string, which was used as input for the learning algorithms.

4 Methodology and Models implementation

4.1 Classical Machine Learning - SVM

The first approach is based on a classical machine learning pipeline that combines TF-IDF vectorization with manually engineered lexical features. This model was designed to serve as a strong baseline while also incorporating domain-specific insights related to clickbait detection.

Vectorization Text data was transformed using a TF-IDF vectorizer with a maximum vocabulary size of 5000 features and an n-gram range of (1,2), capturing both unigrams and bigrams. This representation allows the model to encode local contextual patterns commonly found in clickbait headlines.

Feature Engineering In addition to TF-IDF features, a set of handcrafted lexical features was introduced. These features were derived from an analysis of the most frequent words appearing in clickbait and non-clickbait headlines. For each headline, the following features were extracted:

- the number of words belonging to a clickbait-specific lexicon,
- the number of words belonging to a non-clickbait lexicon,
- headline length (number of tokens),
- number of exclamation marks,
- number of question marks.

Furthermore, a customized TF-IDF weighting strategy was proposed, where terms frequently associated with clickbait were penalized by reducing their inverse document frequency (IDF) values. This modification aims to prevent the classifier from over-relying on superficial clickbait cues.

Classifier The final feature representation consisted of a concatenation between TF-IDF vectors and lexical features. A Support Vector Machine (SVM) classifier with a polynomial kernel of degree 2 was employed for classification.

Hyperparameters The SVM model was trained using a regularization parameter $C = 1.0$. The polynomial kernel was selected after preliminary experimentation, as it provided a good balance between model complexity and generalization performance. A fixed random seed was used to ensure reproducibility.

4.2 Transformer-Based Model - RoBERTa

The second approach is based on a neural architecture built on top of a pretrained transformer model. RoBERTa-base was selected due to its strong performance on text classification tasks.

Architecture The model uses the RoBERTa encoder to generate contextualized token representations. Instead of relying solely on the [CLS] token, a custom attention-based pooling mechanism was introduced. This mechanism computes attention weights over all token embeddings and produces a weighted sum, allowing the model to focus on the most informative parts of the headline.

The pooled representation is passed through a fully connected layer with ReLU activation, followed by dropout regularization and a final sigmoid output layer for binary classification.

While RoBERTa internally employs feed-forward networks within each transformer layer, these operate at token level and do not perform sequence-level pooling. Therefore, we introduce an attention-based pooling mechanism to aggregate token representations before classification.

Technical Contribution The main technical contribution of this model consists of the attention-based pooling layer added on top of the RoBERTa encoder. This layer enables adaptive aggregation of token-level information and provides a more flexible alternative to standard pooling strategies.

Loss Function and Regularization The model was optimized using binary cross-entropy loss. Dropout regularization with a rate of 0.3 was applied to reduce overfitting. Additionally, a weighted loss formulation was explored in order to emphasize correct classification of clickbait headlines.

Fine-Tuning Details All layers of the RoBERTa model were fine-tuned during training. Optimization was performed using the AdamW optimizer with a learning rate of 2×10^{-5} . A linear learning rate scheduler without warm-up steps was employed to stabilize training.

4.3 Experimental Setup

All experiments were conducted using a fixed train/validation/test split of 80%, 15% and 5%, respectively. A fixed random seed was used across all experiments to ensure reproducibility. The validation set was used exclusively for model selection and hyperparameter tuning, while the test set was kept unseen until the final evaluation.

Training Configuration For the transformer-based model, training was performed for 3 epochs using a batch size of 16. The maximum input sequence length was set to 128 tokens. Optimization was carried out using the AdamW optimizer with a learning rate of 2×10^{-5} and a linear learning rate scheduler. All layers of the RoBERTa encoder were fine-tuned during training. Dropout regularization with a rate of 0.3 was applied in the classification head to mitigate overfitting.

For the classical machine learning model, no iterative training process was required. The Support Vector Machine was trained on the combined TF-IDF and lexical feature representation using a polynomial kernel of degree 2.

Evaluation Metrics Model performance was assessed using multiple standard classification metrics. Precision, recall, and macro-averaged F1-score were computed in order to account for both classes equally. Additionally, confusion matrices were analyzed to better understand the distribution of classification errors.

To further evaluate the quality of the probabilistic outputs, precision-recall curves were generated and the area under the curve (PR-AUC) was computed. This metric is particularly informative for binary classification tasks such as clickbait detection, where the balance between precision and recall is important.

5 Results and Evaluation

5.1 SVM Evaluation Metrics

The performance of the classical SVM-based model was evaluated on both the validation and test sets using standard classification metrics. Table 1 summarizes the results.

Table 1: SVM performance on validation and test sets

Dataset	Precision	Recall	Macro-F1	Accuracy	PR-AUC
Validation	0.95	0.78	0.87	0.87	–
Test	0.95	0.78	0.87	0.87	0.97

On the validation set, the model achieved a macro-averaged F1-score of 0.87, indicating balanced performance across the clickbait and non-clickbait classes. A similar performance was observed on the test set, with a macro-F1 score of 0.87, suggesting good generalization and limited overfitting.

The model shows higher precision for the clickbait class, while recall is comparatively lower. This behavior indicates that the classifier is conservative when predicting clickbait headlines, favoring precision over recall. In contrast, non-clickbait headlines exhibit high recall, meaning that most non-clickbait samples are correctly identified.

In addition to threshold-based metrics, the probabilistic outputs of the SVM were evaluated using the precision–recall curve. The model achieved a PR-AUC score of 0.97 on the test set, demonstrating strong ranking capability and reliable probability estimates.

Confusion Matrix Analysis The images illustrates the confusion matrices obtained on the validation and test sets for the SVM-based model.

On the validation set, the model correctly classifies the majority of non-clickbait headlines, with 2296 true negatives out of 2400 samples, resulting in a very high recall for the non-clickbait class. In contrast, clickbait headlines exhibit a higher number of false negatives (533), indicating that a substantial portion of clickbait samples are misclassified as non-clickbait. This behavior confirms that the model prioritizes precision over recall for the clickbait class.

A similar pattern can be observed on the test set. The number of true negatives (767) remains high, while false positives are limited (33), further demonstrating the model’s conservative nature when predicting clickbait. However, 180 clickbait headlines are incorrectly classified as non-clickbait, reinforcing the observation that the model tends to avoid false clickbait predictions at the cost of reduced recall.

Overall, the consistency between validation and test confusion matrices indicates stable generalization. The observed error distribution suggests that the model is more reliable at rejecting non-clickbait headlines than aggressively detecting all clickbait instances, which aligns with the high precision and slightly lower recall reported in the evaluation metrics.

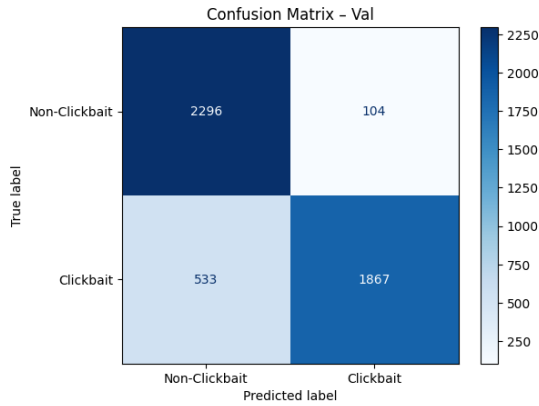


Figure 5: Confusion Matrix – Val

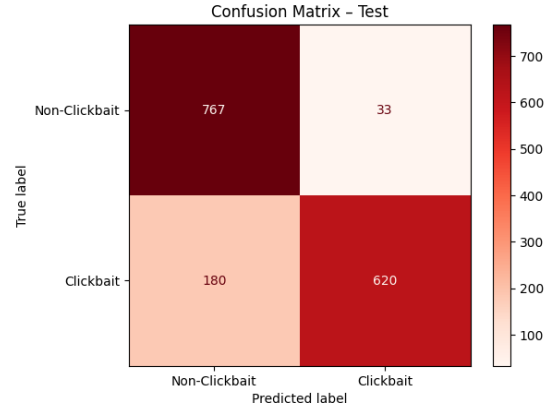


Figure 6: Confusion Matrix – Test

5.2 RoBERTa Evaluation Metrics

The transformer-based model was evaluated using the same protocol as the classical SVM approach, ensuring a fair comparison. Performance was assessed on both the validation and test sets using standard classification metrics.

On the validation set, the RoBERTa-based model achieved a macro-averaged F1-score of 0.97, indicating excellent and balanced performance across both clickbait and non-clickbait classes. Precision and recall values are consistently high for both classes, suggesting that the model is equally effective at detecting clickbait headlines and correctly rejecting non-clickbait content.

Similar results were observed on the test set, where the model achieved a macro-F1 score of 0.97. The close alignment between validation and test performance indicates strong generalization and minimal overfitting. Compared to the classical SVM model, the transformer-based approach demonstrates a substantially improved ability to capture contextual and semantic patterns present in news headlines.

In addition to threshold-based metrics, the probabilistic outputs of the RoBERTa model were evaluated using precision–recall curves. The model achieved a PR-AUC score of 0.996 on the test set, highlighting its excellent ranking capability and confidence calibration.

Table 2: RoBERTa performance on validation and test sets					
Dataset	Precision	Recall	Macro-F1	Accuracy	PR-AUC
Validation	0.97	0.97	0.97	0.97	–
Test	0.97	0.97	0.97	0.97	0.99

Confusion Matrix Analysis The images presents the confusion matrices obtained on the validation and test sets for the RoBERTa-based model.

On the validation set, the model correctly classifies the vast majority of samples from both classes. Specifically, 2317 non-clickbait headlines are correctly identified, with only 83 false positives. Similarly, 2338 clickbait headlines are correctly classified, while only 62 samples are misclassified as non-clickbait. This balanced distribution of errors highlights the model’s ability to handle both classes symmetrically.

The test set exhibits a nearly identical pattern. The model correctly classifies 775 non-clickbait and 779 clickbait headlines, with a very small number of misclassifications (25 false

positives and 21 false negatives). This confirms the robustness of the model and its strong generalization capability.

Overall, the confusion matrices reveal that the RoBERTa-based model achieves both high precision and high recall for each class, significantly reducing the trade-off observed in the classical SVM model. The low and well-balanced number of errors across validation and test sets further supports the effectiveness of the proposed attention-based pooling mechanism.

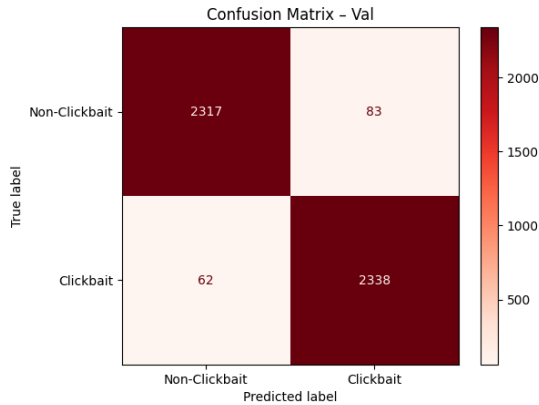


Figure 7: Confusion Matrix – Val

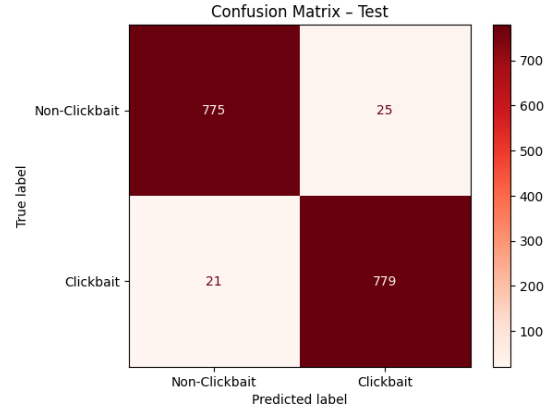


Figure 8: Confusion Matrix – Test

5.3 Comparative Analysis

The comparative results highlight a clear performance gap between the classical machine learning approach and the transformer-based model. While the SVM achieves competitive precision and strong ranking performance, its lower recall for the clickbait class indicates a conservative prediction strategy.

In contrast, the RoBERTa-based model consistently outperforms the SVM across all evaluated metrics. The transformer achieves a substantially higher macro-F1 score and demonstrates a well-balanced precision–recall trade-off, as confirmed by the confusion matrix analysis. This suggests that contextualized representations are more effective at capturing subtle linguistic patterns characteristic of clickbait headlines.

Overall, the results indicate that handcrafted lexical features combined with TF–IDF representations provide a strong baseline, while the transformer-based approach offers superior performance by leveraging deep contextual understanding and attention-based pooling.

Table 3: Comparative performance of SVM and RoBERTa models on the test set

Model	Precision	Recall	Macro-F1	Accuracy	PR-AUC
SVM	0.95	0.78	0.87	0.87	0.97
RoBERTa	0.97	0.97	0.97	0.97	0.99

5.4 Error Analysis

An error analysis was conducted to identify common misclassification patterns and to better understand the limitations of the evaluated models. The analysis focuses on error types, representative cases, and their implications in real-world scenarios.

Types of Errors The SVM-based model exhibits a clear tendency toward false negatives for the clickbait class, misclassifying a considerable number of clickbait headlines as non-clickbait while maintaining a low number of false positives. This behavior reflects a conservative prediction strategy that favors high precision over recall. In contrast, the RoBERTa-based model produces significantly fewer errors and shows a more balanced distribution between false positives and false negatives.

Representative Error Patterns Inspection of misclassified samples indicates that the SVM model struggles with subtle or implicit clickbait headlines that do not rely on explicit lexical markers. Conversely, occasional false positives occur when factual headlines contain words commonly associated with clickbait. The few errors made by the RoBERTa model typically correspond to borderline cases that combine informative and sensational elements, where the distinction between clickbait and non-clickbait is inherently ambiguous.

Interpretation and Real-World Implications In real-world scenarios, the conservative nature of the SVM model may be advantageous in applications where minimizing false clickbait detections is important, such as editorial support tools. However, its lower recall limits its effectiveness for comprehensive clickbait detection. The RoBERTa-based model, with its balanced error profile and strong contextual understanding, is better suited for deployment in practical systems that require both high precision and high recall.

Overall, this analysis highlights the trade-off between classical and transformer-based approaches and supports the use of deep contextual models for robust clickbait detection in real-world applications.

6 Critical Analysis

This section provides a comparative interpretation of the experimental results, discusses the limitations of the proposed approaches, and highlights potential sources of bias that may affect model performance in real-world scenarios.

6.1 Comparative Analysis

The experimental results reveal a clear performance difference between the classical SVM-based approach and the transformer-based RoBERTa model. While the SVM achieves high precision and strong ranking performance, its lower recall for the clickbait class indicates a conservative prediction strategy. This behavior suggests that the model is effective at avoiding false clickbait detections but fails to capture more subtle or implicit clickbait patterns.

In contrast, the RoBERTa-based model demonstrates consistently higher and more balanced performance across all evaluation metrics. Its ability to leverage contextualized representations allows it to capture nuanced linguistic and semantic cues that are not easily modeled using sparse lexical features. As a result, the transformer-based approach is better suited for practical deployment scenarios where both precision and recall are critical.

6.2 Limitations

Despite the strong empirical results, several limitations should be acknowledged. First, the dataset used in this study is not fully representative of contemporary online news content. Click-

bait strategies and headline styles have evolved over time, with recent headlines increasingly relying on subtle framing, irony, or topic-specific trends. Consequently, model performance may degrade when applied to more recent or domain-shifted data.

Additionally, the SVM-based model relies heavily on handcrafted lexical features and TF-IDF representations, which limits its ability to generalize beyond the vocabulary and patterns observed during training. Although the transformer-based model mitigates this issue to some extent, it remains sensitive to distributional shifts in language use and topic coverage.

6.3 Bias and Real-World Implications

Potential sources of bias stem primarily from the dataset composition and labeling process. Certain topics or writing styles may be overrepresented, leading the models to associate specific lexical patterns with clickbait or non-clickbait labels. This bias can result in systematic missclassification of headlines from underrepresented domains.

In real-world scenarios, such biases may affect the fairness and reliability of automated clickbait detection systems, particularly when deployed across diverse news sources or cultural contexts. Addressing these issues would require more recent and diverse datasets, as well as continuous model adaptation to evolving headline styles.

Overall, this analysis highlights the importance of combining strong quantitative performance with critical evaluation when deploying clickbait detection models in real-world applications.

This section provides a comparative interpretation of the experimental results, discusses the limitations of the proposed approaches, and highlights potential sources of bias that may affect model performance in real-world scenarios.

7 Conclusions

This paper investigated the problem of automatic clickbait headline detection by comparing a classical machine learning approach with a modern transformer-based model. Two systems were implemented and evaluated: a Support Vector Machine (SVM) classifier using TF-IDF and handcrafted lexical features, and a RoBERTa-based neural model enhanced with an attention-based pooling mechanism.

Experimental results demonstrate that both approaches are effective for clickbait detection, but with notable differences in performance and behavior. The SVM-based model achieved strong precision and ranking performance, serving as a robust and interpretable baseline. However, its lower recall for the clickbait class indicates a conservative prediction strategy, leading to missed clickbait instances.

In contrast, the RoBERTa-based model consistently outperformed the SVM across all evaluation metrics, achieving higher accuracy, macro-F1 score, and PR-AUC. Its balanced precision-recall trade-off and low error rates confirm the advantage of deep contextual representations for capturing subtle linguistic patterns characteristic of clickbait headlines.

The comparative analysis confirms that while traditional machine learning methods combined with feature engineering can provide competitive results, transformer-based models are better suited for complex and evolving language phenomena such as clickbait. The inclusion of an attention-based pooling mechanism further enhances the model’s ability to focus on informative parts of the headline, contributing to improved robustness and generalization.

Overall, this work highlights the importance of contextual modeling in modern NLP applications and demonstrates that deep learning approaches offer a more reliable solution for

clickbait detection in real-world settings.

7.1 Future Work

Several directions for future work can be explored. One promising extension of this project is the development of a Google Chrome browser extension that automatically scans web pages, extracts news headlines, and identifies clickbait content in real time. Such an extension could present users with a filtered list of detected clickbait headlines or visually highlight them, helping users make more informed content consumption decisions.

Additionally, future work could involve training and evaluating the models on more recent and diverse datasets to better capture evolving clickbait strategies. Incorporating multilingual data, multimodal features (such as images), or user interaction signals may further improve detection performance. Continuous model updating and domain adaptation techniques would also be essential for maintaining accuracy in dynamic online media environments.

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